

A Thematic Analysis on the Lack of Visual Mental Imagery

Dániel Péntzes

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Introduction

The history of investigating the form of imagination is quite extensive, expanding from philosophers of ancient times up to recent findings in psychological experiments. The word ‘phantasia’ originates from ancient Greece, meaning “to bring to light”, “to appear”, or “to make appear...” (p21, Brann, 1991), and it is a term mostly understood for imagination. In 2015 however Zeman, Dewar, and Della Sala coined the term ‘aphantasia’, signifying the concept that there are people who lack one specific form of imagination: voluntary visual mental imagery¹. This study received a considerable amount of media attention, whereby e.g. Facebook groups were created for those who claimed to be ‘aphantasic’². Although psychology has more than a 100-year-old tradition in investigating individual differences in mental imagery (see e.g. Galton, 1883, cited in Brann, 1991), the main paradigm in the field rests on quantitative research. Qualitative research seems to be scanty. On the one hand, clinical psychology employs interviews to understand patients’ experiences with respect to their mental disorders (e.g. Hackmann, Surawy, Clark, 1998; Speckens, Hackmann, Ehlers, Cuthbert, 2007). On the other hand, and more importantly for the current purposes, only a handful of qualitative studies aim at analysing the experiences of people who lack mental imagery (e.g. Watkins, 2018; Ganczarek, Żurawska-Żyła, & Rolek, 2020). Therefore, this study employs a qualitative approach with respect to understanding one “specific” form of mental imagery: the absence of its visual properties.

Due to this study being qualitative, interpretivism and constructionism will form the basis of this paper (Chapter One by Ormston, Spencer, Barnard, & Snape, in Ritchie, Lewis, McNaughton Nicholls, & Ormston, 2014). The former roughly equals to the idea that our social constructs are not set in stone; we as humans constantly interpret the experiences we observe. The latter further accentuates that humans actively create their own meanings. Therefore, the goal of the current qualitative approach is to understand how the form (or the lack) of our mental imagery is created and interpreted.

To further emphasize the interpretative approach, the transparency hypothesis will be borrowed (Kind, 2010). According to transparency our perceptual experiences are transparent, that is, when we look at e.g. an apple, our attention is on the object of our experience but not on the experience itself. This supports the idea that we cannot fully grasp the phenomenal aspects of our perceptual experiences. Since mental imagery is usually understood as having perceptual qualities (Brann, 1991), the transparency hypothesis can be applied to visual mental imagery as well. Therefore, it can be argued that when one is having a mental image, the attention is on the *content* of the mental image but not on the *form* of the experience itself. The implication is that one will be interpreting its own mental imagery experiences. For example, propositional attitudes (Kind, 2010) about our experiences may differ. My belief e.g. is that the visual aspects of my own mental imagery are quite detailed, therefore I claim that my visual mental imagery is intact. However, a person with aphantasia might claim that there are no visual aspects to her mental imagery, therefore she claims to have no visual mental imagery at all.

To this end, the main research question is as follows: How do people with self-ascribed aphantasia experience and understand the lack of their visual mental imagery? To better understand this point of view a semi-structured interview was conducted as a data collection method (Chapter Seven by Yeo et al., in Ritchie et al., 2014). The epistemological approach of the interview was of a cartographer mapping out an uncharted territory (ibid., p253). That is, the interviewer’s questions aimed at mapping out the interviewee’s interpretations and experiences. This was then followed by a thematic analysis (Braun & Clarke, 2006) to find common, underlying themes in interpretations. This analysis is suitable within the constructionist framework. Since the data will be interpreted through the transparency

¹ The lack of mental imagery is not a new phenomenon, see Brann (1991) for an elaborate historical overview.

² See e.g. <https://www.facebook.com/groups/204603509580186/>

hypothesis (Kind, 2010), a deductive or theory-driven approach was employed (Braun & Clarke, 2006).

Data Collection

Participants were recruited from the aforementioned Facebook group (see footnote 2) via a public announcement to which people with self-ascribed aphantasia responded. Four people were then selected on a “first come first served” basis, all women, and between the ages of 38 and 67. Three of them were native English speakers. As the study’s concern was to investigate people’s experiences and interpretations, no objective measures were taken to “verify” the lack of mental imagery, such as the Vividness of Visual Imagery Questionnaire used by Zeeman et al. (2015).

A consent form was first provided via an introductory email, assuring voluntary participation, a right to cancel participation, and anonymity in data analysis and in reporting (Chapter Four by Webster, Lewis, & Brown, in Ritchie et al., 2014). Participants either sent it back in a written form or provided verbal consent at the beginning of the interview, which was conducted via Zoom (Zoom Video Communications Inc., USA) and lasted about 30 minutes. These online sessions were recorded and were auto transcribed by Otter (Otter.ai, USA). All documents, including signed consent forms, video files, and transcriptions were saved to a folder created by Umeå University, and were deleted from local computers.

The structure of the interview is amalgamated from several sources. After the introduction and warm-up questions, it is partitioned to three main parts following Hackmann et al. (1998): general experiences (e.g. “When someone asks you to visualize something, what is your experience?”); the last experience (e.g. “Can you recall the last time you experienced any forms of mental imagery?”); and a specific task. This latter part used two examples from existing literature; first, participants were asked to think about the main entrance of their house and describe it (Schwitzgebel, 2002); second, participants were asked to think about the breakfast they had and describe it (Galton, 1883, cited in Brann, 1991). Lastly, other forms of mental imagery (e.g. auditory) were queried, and the interview was closed with any remaining questions the participants had³. This threefold structure is useful, for one it probes the participant to formulate their general beliefs on their experiences, yet it also examines ad hoc cognitive tasks that could relate to mental imagery experiences.

Data Analysis

As per Braun and Clarke (2006), the following steps of a deductive thematic analysis were conducted: the four interview transcripts were first read through separately. Then, individual codes were identified within each transcript that covered one specific topic related to mental imagery experiences. These codes were then compared between the transcripts, to look for bigger themes not just including one person’s experience. Given the deductive approach, the initial themes were searched for with respect to how participants described a lack of mental imagery experience, furthermore how this might relate to the transparency hypothesis (Kind, 2010). Following this, themes were revised; additionally, it was decided whether they are descriptive (closer to the participants’ experiences) or interpretative (closer to the researcher’s interpretations). A final list of themes, including codes, can be found in Appendix B. Altogether, 11 themes and 48 codes were identified.

³ The whole structure of the interview can be found in Appendix A.

Findings

The large number of themes might indicate a disadvantage, for it can always be the case to combine more of them together. It was however decided to keep them as such, for they show a clear linearity in the complex narrative of how one might experience and understand a lack of visual mental imagery. Therefore, each paragraph will showcase a theme (with the theme's name in *italics*), including a detailed description, further codes and quotes, and possible interpretations.

No Visualization / No Pictures. Participants reported on not having any visual qualities to their inner experiences. They do not see "pictures", nor they can visualize anything if asked to do so.

Eyes Closed: Black. Participants reported on having a blackness when closing their eyes and not having any visual mental imagery. This is usually contrasted with others' experiences who might not have this blackness upon closing the eyes. As one of the participants described it: "it's just looking at the inside of my eyelids".

No Mental Imagery in Any Modalities. Participants reported on lacking not just visual mental imagery, but most of the time also auditory, olfactory, or any other type of mental imagery. There is no apparent experience of imagining sound, smell, touch, or even taste. One of the participants put it as such: "I have nothing". This might become apparent when one is reading (e.g. not having an inner speech narrating the text), or when one is thinking about the taste of her last meal.

Thinking of / Knowing. Participants did report on thinking about things, or knowing about things, as opposed to having visual experiences. This expression of "thinking of" was quite emphasized and marked something important in their experiences, for example: "I just know I am thinking of an (apple)". Consequently, how one knows it or how one thinks of it seemed to be hard to describe; this is just something participants knew and do most of the time. This might invite the possibility of the transparency hypothesis (Kind, 2010), for it seems difficult (if not impossible) to describe the *form* of the experience of "thinking of" as opposed to concentrating on its *contents*.

Not Having Mental Imagery as Normal. Participants reported on never having mental imagery throughout their whole life. Sometimes they described this as a condition since their birth. Subsequently, they always considered this as a normal state. It is important to emphasize the experience being given in one's life (as the personal norm), since this is what will be challenged by the following theme.

Not Knowing About Others' Experiences. Participants did not know that others have visual qualities to their mental imagery, until they found out from Facebook or through other media attention, following the study of Zeeman et al. (2015).

Surprise of Finding Out. Participants were surprised upon finding out that others have mental imagery. Some of them "couldn't believe", others commented that it "blew her mind". Following the aforementioned publication, most participants started conversations with family members, friends, and even started a discourse with other people who similarly did not experience any mental imagery. Subsequently, participants were amazed of those who have mental imagery, something described as being a "magical" thing, or a "superpower".

Others Seeing or Watching Mental Imagery. Participants reported that others can see or even watch mental imagery in their minds; something they lack of. Either others "see pictures in their mind", or it is "like watching a movie". This became most apparent when participants had a conversation with people who mentioned not having that blackness upon closing the eyes, but rather vivid visual mental imagery.

Curiosity. This even drives them to find out more about (the lack of) mental imagery. This for example manifested in the previously mentioned Facebook group, where people with (or without) mental imagery share their experiences and start conversations.

Why not having mental imagery is good?. Three participants reported that not having mental imagery can also be advantageous, based on their life experiences. For example, two of them described that this way it is easier to emotionally detach oneself from a traumatic experience (e.g. not remembering the image of a family member's death, or e.g. in an emergency situation one can focus better without imagining any horrible scenarios). Another participant described that by not having any mental imagery one can be socially more attentive; since she is not "distracted by images and sounds in my mind" she is able to listen more to the other person. Apparently, not having mental imagery might be even useful in some cases.

The Imagery Tasks - Interpretations & Transparency. The themes so far were descriptive, insofar as they described a topic coming straight from the participants. However, the two imagery tasks provided an opportunity to employ an interpretative theme as well. The transparency hypothesis (Kind, 2010) might have become manifest in how participants reported their experiences; seemingly they rather focused on the contents as opposed to the forms. Both scenarios (the main entrance and the breakfast) included naming places (e.g. porch, door, kitchen), materials (e.g. metal, plastic, wood), or verbs (e.g. open, pull, cut). Answering to the question how these experiences differ (or not) seemed to be difficult; it either depended on memory, on interest (buildings versus foods), or even whether the two events happen in different spaces (outside versus inside). Whether it was about "thinking of" them or "imagining" them was hard to pinpoint; a question also surfacing in the fact that participants reported having colours, which would indicate perceptual aspects. Admittedly, even the author of this paper has a hard time describing the forms of these experiences.

Concluding Remarks

Overall, it can be claimed that the interviewed people always considered themselves not having any visual (or any other) mental imagery experiences, and for them this was a given. Most of the time they rather emphasized thinking about concepts instead of perceptually formulating them. Finding out about other people's experiences, it hit them as a surprise, and an amazement over the possibility that other people do have different, perceptual, qualities to their inner experiences. This also led them to be curious to find out more about other people. Furthermore, a lack of mental imagery could also be interpreted as an advantageous fact.

Certain individual elements (or codes) have been left out intentionally. Some of them described their dreams, which although might have contained visual elements, there was no common agreement on how one can remember them. Some participants also shared other life experiences that for example were related to emotional events. While these experiences certainly have their importance to their beholders, they were not strictly related to the lived experience of a lack of mental imagery.

An interesting question was left open whether the transparency hypothesis (Kind, 2010) would be applicable in interpreting these experiences. While the author of this paper believes this theory to be one explanation, it cannot be claimed that support was found for it. For example, it could have always been the case that during the imagery tasks the questions formulated by the interviewer were understood in a different way than they were intended to. During the interviews we did not go into details over comparing explicitly the forms of the two scenarios (the main entrance and the breakfast), and perhaps more probing questions would have led to more clarity. It could also be the case that fitting 14 questions to 30 minutes led to overlooking important details. However, the author of this paper believes that using such a theoretical framework could prove fruitful in future studies, as to how to differentiate a form of an inner experience from its contents.

Further limitations of this study worth mentioning. Given that the interviews were taken with four people and given the interpretative and constructionist framework of qualitative designs, generalizing these findings to a bigger population cannot be done. This would only be applicable in a quantitative framework where controlled measures and precise statistical analyses can provide tools for generalizations (Chapter One by Ormston et al., in Ritchie et al., 2014). Furthermore, there is a limitation on how the interviewer deals with his semi-structured questions during the interviews. Demand characteristics can always show up during the discussions such that the interviewer's questions may alter the interviewee's perceptions or interpretations (Schwitzgebel, & Hurlburt, 2007). For example, I asked the participants to "think" about either the main entrance or the breakfast; I intentionally did not use the word "imagine" not to imply an expectation of a mental image. This however could have always altered the way they experienced the two scenarios, or the way they interpreted them.

What is important to note here is the differentiation between what might happen in an individual experience and how one might interpret it. Schwitzgebel (2002) argues that knowing our visual mental imagery experiences is inherently difficult. Describing all the fine details of thinking about (or imagining) our front door is a challenging task. This does not imply however that the individual experience is not true for its beholder. Understanding the totality of another person's experiences might be an immense enterprise, but at least we have some tools to begin with. As such, the applied thematic analysis of this study offered one option to have a better grasp on how the lack of mental imagery might appear to some.

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Appendix A

Interview Structure

INTRO:

- 1) Do you consider yourself as someone not having visual mental imagery?
- 2) What aspects of mental imagery do you consider yourself missing in your experience?

HISTORY:

- 3) Since when do you consider yourself as not having mental imagery?
- 4) How did you find it out? Can you describe the situation?
- 5) What did you think when you found it out? What was your reaction?
- 6) How do you think people experience their visual mental imagery? What do you find most puzzling about other people's descriptions?

GENERAL EXPERIENCE:

- 7) When someone asks you to visualize something, what is your experience? Can you describe it?

THE LAST EXPERIENCE:

- 8) Can you recall the last time when you experienced mental imagery? What was the situation?

IMAGERY TASKS:

- 9) Here is an example of a situation, and I would like to ask you to think about them and describe what you experience.

10) Think about the front door of your house or your building, the main entrance. Can you describe it to me?

11) Recall your breakfast this morning. Can you describe it to me?

- 12) “In what way is this experience (your breakfast this morning) the SAME OR DIFFERENT than the previous experience (the entrance of your building)?”⁴

OTHER:

- 13) Apart from visual mental imagery, what other forms of experience do you use/have?

E.g. Auditory, more abstract/thought-like, taste, smell, etc.

- 14) Is there anything else you'd like to add?

⁴ This question comes straight from Schwitzgebel, & Hurlburt (2007), for it is concerned as “one of the most non-leading questions possible.” (p112).

Appendix B

Themes & Codes

<i>No Visualization / No Pictures</i> No Visualization "no pictures" only a problem when somebody asks to visualize	<i>Eyes Closed: Black</i> "Totally black" "when I close my eyes I can see only darkness" "black screen" when eyes closed closed eyes: black, "nothing" "looking at the inside of my eyelids" "black" only light if, "black" "I can visualize but it's all dark" She can imagine "but it's on the dark"	<i>No Mental Imagery in Any Modalities</i> "zero mental imagery" No Mental Imagery of Taste, Touch, Sound, Feel, Smell, See "I have nothing" None of the senses have no hearing, touch, taste
<i>Thinking of / Knowing</i> she can "think of something" "I just know I am thinking of an ..." If asked she explains what she knows about it she can think of it she knows how it looks like She can remember, think of it, but no senses at all "I can think of words"	<i>Not Having Mental Imagery as Normal</i> No Mental Imagery from "Birth" "I've never been able to see stuff" Used to Consider Not Having Mental Imagery as "Normal" She never realized this "I never thought that was a thing" just "a word that people use" it was always like that, so it was not a loss	<i>Not Knowing About Others' Experiences</i> Not Knowing that People "see" mental images Did not know others have mental imagery
<i>Surprise of Finding Out</i> Others have "these superpowers" "Couldn't believe" "amazing" "magical" "blew her mind" that others have mental imagery	<i>Others Seeing or Watching Mental Imagery</i> People "see pictures in their mind" His husband can see "that he remembers or imagines" "surprise": closed eyes and see things "people could actually see it because that's just not my experience" "Everyone in my family can visualize everything" Others can do it Others: literally watching a video "in their head" "like watching a movie"	<i>Curiosity</i> "Curious" "curious" "to learn more about the limits of my abilities"
<i>Why not having mental imagery is good?</i> No mental image, therefore emotionally detach from experiences More attentive to others because "I'm not distracted by images and sounds in my mind" Since no visual mental imagery, no trauma	<i>The Imagery Tasks - Interpretations & Transparency</i>	